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Primes in $\mathbb{Z}[i]$

Lem. Every prime in $\mathbb{Z}[i]$ divides exactly one prime number.

Note: For $a, b \in \mathbb{Z}$,

$$\gcd(a, b) = 1 \text{ in } \mathbb{Z} \iff \gcd(a, b) = 1 \text{ in } \mathbb{Z}[i].$$

Thm (Primes in $\mathbb{Z}[i]$). Let $\pi \in \mathbb{Z}[i]$ be prime.

1. $\pi = \pm 1 \pm i \mid 2$.

2. $\pi = p \equiv 3 \pmod{4}$ or one of its associates

3. $\pi = x + iy \mid p$ with $x, y \neq 0$, where $p \equiv 1 \pmod{4}$.

In fact,

$$N(\pi) = \pi \bar{\pi} = x^2 + y^2 = p.$$

If π is such a prime divisor of p , there are

8 prime divisors of p in total:

$$\pm \pi, \pm i\pi, \pm \bar{\pi}, \pm i\bar{\pi}.$$

So there are exactly 8 ways of writing $p = x^2 + y^2$.

pf. Let $\pi \in \mathbb{Z}[i]$ be prime. There is a unique prime number p s.t. $\pi | p$. Say $\pi = x + yi$. Then

$$N(\pi) = x^2 + y^2 \mid N(p) = p^2.$$

So

1. $N(\pi) = p^2$: π is an associate of p , hence p is prime in $\mathbb{Z}[i]$.

2. $N(\pi) = p$.

(1) $p = 2$: $N(\pi) = x^2 + y^2 = 2$. Solving this gives

$$(x, y) = (\pm 1, \pm 1).$$

These corresponds to $\pi = \pm 1 \pm i$. Clearly,

$$2 = (1+i)(1-i),$$

so 2 is not prime in $\mathbb{Z}[i]$.

$$(2) \quad p \equiv 3 \pmod{4}.$$

Want to show p is prime, i.e., any divisor

α/p is either a unit or an associate.

Assume α/p is not a unit or an associate. Then,

with $\alpha = z + w i$,

$$N(\alpha) = z^2 + w^2 = p.$$

Since $z^2, w^2 \equiv 0$ or $1 \pmod{4}$, $z^2 + w^2 \equiv 0, 1$ or 2

$\pmod{4}$, a contradiction. This shows p is prime.

Recall $\pi | p$. So $\pi = p$ or an associate of p .

$$(3) \quad p \equiv 1 \pmod{4}.$$

Want to show p is not prime.

Since $-1 \not\equiv 0 \pmod{p}$, $\exists x \in \mathbb{Z}$ s.t. $x^2 \equiv -1 \pmod{p}$.

So $x^2 + 1 \equiv 0 \pmod{p}$, i.e.,

$$p | x^2 + 1 = (x+i)(x-i).$$

By Euclid, $p | x+i$ or $p | x-i$. But neither can happen because

$$\frac{x \pm i}{p} = \frac{x}{p} \pm \boxed{\frac{1}{p}} i \notin \mathbb{Z}[i].$$

a contradiction. So p is not prime in $\mathbb{Z}[i]$.

Thus

$$N(\pi) = \pi \bar{\pi} = (x + yi)(x - yi) = x^2 + y^2 = p.$$

Because $\mathbb{Z}[i]$ is a UFD, any prime factor of p must be an associate of π or $\bar{\pi}$.

Moreover, p is a prime number $\Rightarrow x, y \neq 0$

$$|x| \neq |y|.$$

So the associates of $\pi = x + yi$:

$$x + yi, -x - yi, -y + xi, y - xi,$$

which are distinct, and the associates of $\bar{\pi} = x - yi$:

$$x - yi, -x + yi, y + xi, -y - xi.$$

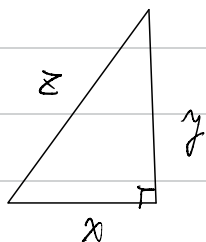
which are distinct, are pairwise distinct. So there are exactly 8 prime factors of p , given by π , $\bar{\pi}$ and their associates.

As a consequence, there are exactly 8 ways of expressing p as a sum of two squares:

$$p = \underbrace{(\pm x)^2 + (\pm y)^2}_{4 \text{ choices}} = \underbrace{(\pm y)^2 + (\pm x)^2}_{4 \text{ choices}}$$



Pythagorean triples.



$$x^2 + y^2 = z^2, \quad x, y, z \in \mathbb{Z}.$$

Eg. $(x, y, z) = (3, 4, 5), (5, 12, 13).$

Goal: Find all $x, y, z \in \mathbb{Z}$ satisfying

$$x^2 + y^2 = z^2.$$

Observation: If (x_0, y_0, z_0) is a solution, then

$(x_0 d, y_0 d, z_0 d)$ is also a solution, where $d \in \mathbb{Z}$.